

Day 8 Homework: automated checks for my chatbot

Mithran Sadasivam · AI Engineer Program, Phase 1 · Week 3, Day 8

I used my own chatbot from Day 7 (the Northwind IT help desk bot, 23 documents, 62 chunks, collection **my_docs**, RAG_MODE=full). I did all the work through Claude Code and then went through the results myself. Everything is in the HW 08 folder and committed in git.

A quick note on setup: the class files had three functions still marked TODO (the citation check, the judge call and the quality gate), so I had those filled in first. All 140 unit tests pass. My test questions file is my 10 questions from Day 6, with each one marked as "answer" or "decline". The browser helper file from the homework did not exist in my folder, so I made a small one with the same names (ask and TIMEOUT).

Task 1: my labels

I asked the chatbot my 10 questions and ran only the code checks first. Both code checks came out at 100% (citations 1.0, behaviour 1.0) and the average time was 10.4 seconds per answer. Then I read every answer next to the source text it cited and labelled it before I ran the judge. The file is data/my_labelled_answers.csv.

#	Question	Correct	Grounded	Why
1	How do I set up MFA on a new phone?	yes	yes	The four steps are word for word in source [1]. The backup codes point is in [2].
2	How long do you keep daily server backups?	yes	yes	30 days, and both [1] and [2] say it.
3	My VPN keeps cutting out at home	yes	yes	Update, cable and TCP are in [1], the 12 hour limit is in [2], the log file advice is in [3].
4	What laptop does a new starter get?	yes	yes	14-inch plus headset and dock in [1], floor 2 in [2], 4 years in [3], extra equipment and 5 days in [4].
5	I forgot my password	yes	yes	Reset page and MFA step in [2], the manager check in [1].
6	How big a file can I email?	yes	yes	25 MB and the NorthDrive link are in [1].
7	I lost my phone, who do I call?	yes	yes	Extension 4357 is in [1], the security email address is in [2].
8	Salary band for a senior engineer?	yes	yes	It should decline and it did. A decline does not claim anything.
9	Which printer on floor 3?	yes	yes	The documents only cover floors 1 and 2, so declining was right.
10	Does Northwind support Linux laptops?	yes	yes	Not in the documents, it declined.

I was surprised that I had no "no" at all. I checked each fact one by one and every sentence with a fact had a citation and the cited source really said it, so I could not honestly mark any of them down.

Task 2: how far can I trust the judge?

Run	Agreement: correct	Agreement: grounded	What changed
My 10 answers, before	10 / 10	10 / 10	nothing
My 10 answers, after	10 / 10	10 / 10	one rubric line added
Class sample (12 answers), before	12 / 12	10 / 12	nothing
Class sample (12 answers), after	12 / 12	11 / 12	same rubric line

On my own answers the judge agreed with me every time, so there were no disagreements to read and nothing to improve. Since I could not show a before and after on that, I also ran the judge on the 12 answers from the class file, which have human labels and some bad answers in them. The one change I made to the judge prompt was adding this line: "Every sentence with a fact needs its own citation."

The two disagreements on the class sample:

1) Answer 4 had no citation at all. The judge said grounded, but the label says not grounded. I think the label is right, because a fact with no citation cannot be checked. My new line did not fix this one, the judge still accepted it on both runs. The code check ("no citations") does catch it, so I would rely on that. A stronger line, like "no [n] in the answer means not grounded", is what I would try next.

2) Answer 12 was a decline. The judge first said not grounded, which is wrong because the rules say a decline is grounded. After my change it agreed with the label, and it stayed that way when I ran it again. To be honest, my line does not mention declines, so I can't say it was the cause. It might just be the judge being a bit inconsistent.

Decision: I kept the change because grounded agreement went from 10/12 to 11/12 (same result on a repeat run) and my own 10/10 did not get worse. What I take from this: the judge's "correct" verdicts looked reliable on these documents, but I should check its "grounded" verdicts against the code checks, and measure again whenever I change the judge prompt or the model.

Task 3: quality gate

I set my thresholds from my Day 7 results and the numbers above. The file is `data/my_quality_thresholds.json`.

Threshold	Value	Why this number
min_hit_rate	0.95	Day 7 measured 100% with full mode on 20 questions (vector was 90%). One question is 5 points, so 0.95 allows one miss for noise.
min_citation_pass_rate	1.0	A broken or missing citation is never acceptable, nobody could check the answer.
min_behaviour_rate	0.9	I measured 10/10 (7 answered, 3 declined). One miss is allowed.
min_judge_correct_rate	0.9	I measured 1.0 and the judge agreed with my labels 10/10.
min_judge_grounding_rate	0.9	I measured 1.0. The judge is weaker on grounded, so the code citation check backs it up.

max_avg_seconds	15	My full mode takes about 10 seconds per answer. The template says 6, but that would fail my bot every time. 15 seconds was the limit I chose on Day 7 for a chat.
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Run	Result	Exit code	Numbers
RAG_MODE=full	QUALITY GATE: PASS	0	hit_rate 1.0, citation 1.0, behaviour 1.0, judge correct 1.0, judge grounded 1.0, 9.83 s
--mode vector	QUALITY GATE: FAIL	1	hit_rate 0.9, needs >= 0.95. Everything else was fine (3.83 s).

Vector fails for a good reason. It is quicker but it missed the right chunk on 2 of the 20 retrieval questions, which matches what I saw on Day 7 (it struggles with follow-up questions). That is exactly what I want the gate to catch if someone switches the mode to save time.

Task 4: browser test

Scenario: someone asks about an error code, and the error-code reference has to show up as a source. This is something users rely on, because error codes are exactly what people paste into the chat. The test is in tests/e2e/e2e_my_checks.py:

```
def test_an_error_code_question_cites_the_error_code_reference(page: Page, app_url: str):
    ask(page, app_url, "What does error VPN-809 mean?")

    expect(page.get_by_text("Day07_Slide05_kb_error_codes.pdf").first).to_be_visible(timeout=TIMEOUT)
    expect(page.get_by_text("DECLINED")).to_have_count(0)
```

It checks the structure (the source tag is on screen and the bot did not decline), not the exact wording of the answer.

Run	Result	What happened
Passing run	1 passed	pytest tests/e2e/e2e_my_checks.py gave "1 passed".
Broken on purpose	1 failed	I deleted the error-code PDF from data/my_docs and rebuilt the index (54 chunks instead of 62). The bot answered from the VPN guide instead, and the test failed with "Locator expected to be visible ... element(s) not found".
Undone	1 passed	I restored the PDF with git checkout, rebuilt the index (back to 62 chunks) and the test passed again.

I also added a rule to CLAUDE.md: run pytest -q after every change, and before tagging a release run the quality gate and the browser tests, and do not tag if either one fails.

Where things stand

140 unit tests pass, the gate passes in full mode, the browser test passes, and everything is committed. The browser test needs Ollama running because it uses the real app and the real answer model.